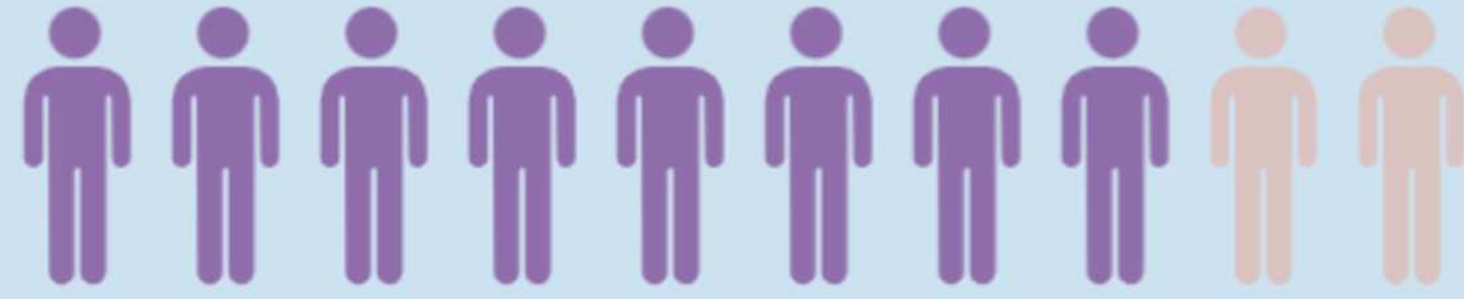


THE EFFECTS OF GLUTAMIC-PYRUVIC TRANSAMINASE-TARGETING siRNA ON HEPATOCELLULAR CARCINOMA AS MODELED IN HEPG2 CELLS

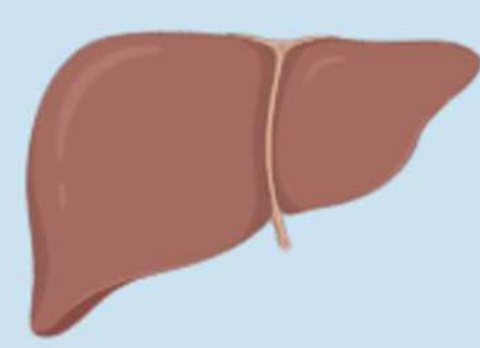
Background

Hepatocellular carcinoma (HCC) is an aggressive cancer, characterized by the **overexpression of ALT** (Guo et al., 2020)

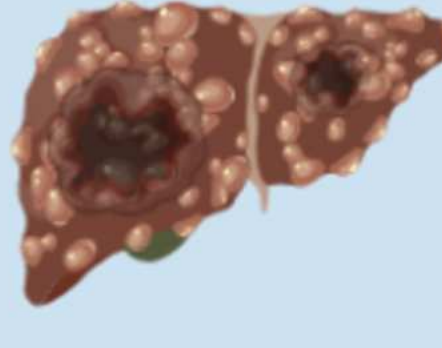


Graphic created by researchers using Biorender, 2026.

8 in 10 HCC patients die from **large tumor formation, severe liver dysfunction, and cirrhosis (scarring)** (Samant et al., 2021)



Healthy liver



HCC-affected liver

Graphic created by researchers using Biorender, 2026.

Overexpression
of ALT

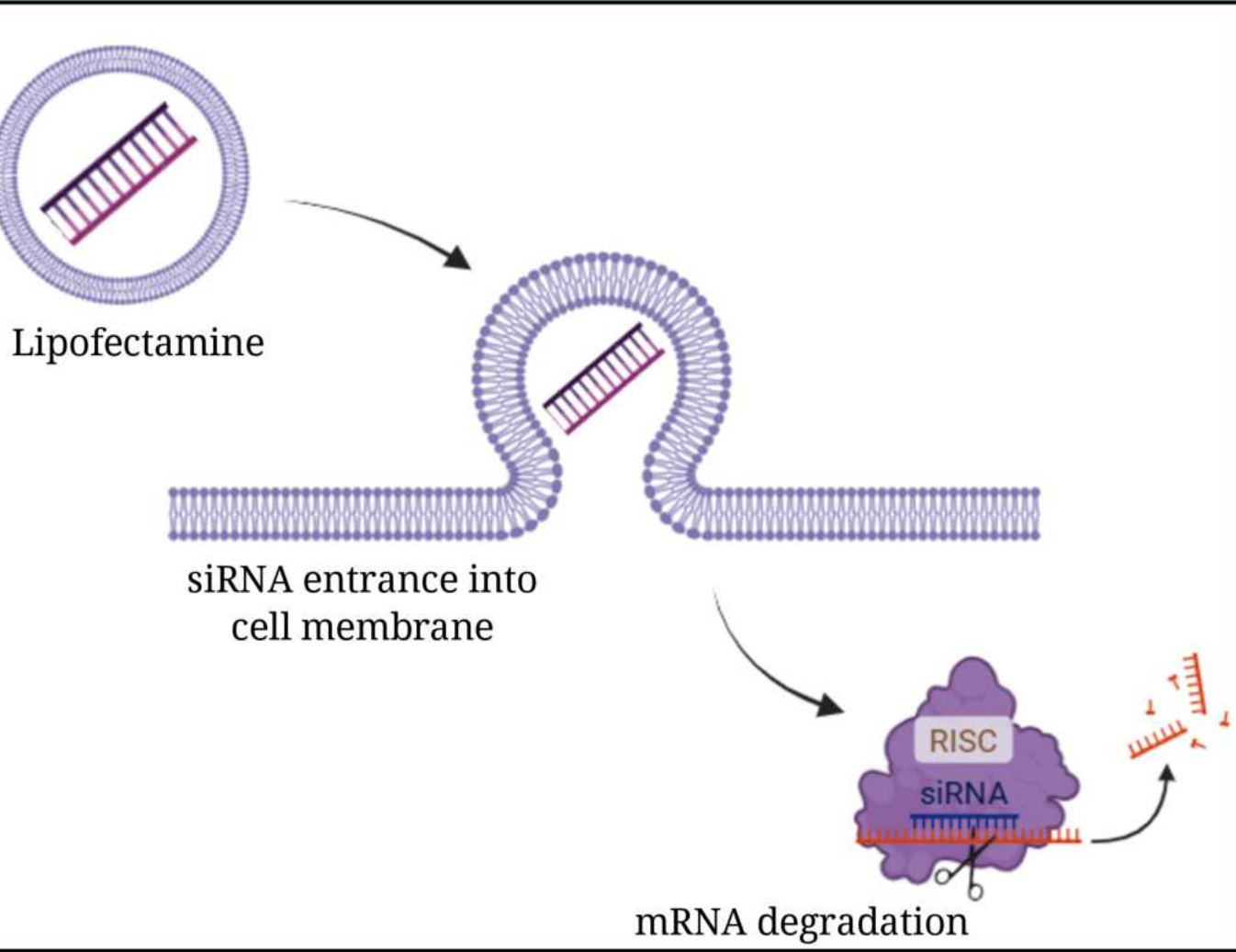
Inflammation
and tumor
formation in
liver cells (HCC)

Alanine Aminotransferase (ALT)

is a protein found primarily in the liver coded by the **Glutamic-Pyruvic Transaminase (GPT)** gene (Fu et al., 2022)

GPT-targeting siRNA

cleaves ALT-encoding mRNA, which should **decrease ALT expression** and overall HepG2 tumor formation (Alshaer et al., 2021).



Graphic created by researchers using Biorender, 2026.

Significance

The purpose of this novel treatment is to offer an alternative to invasive and demanding current methods of treatment by **suppressing the expression of ALT**.

Hypothesis

If HepG2 cells are transfected with GPT-targeting siRNA, then **ALT protein will decrease** and **cell viability will decrease**. siRNAs silence the GPT gene, so the production of ALT will be suppressed. With these crucial pathways impacted, the cancerous cells will be less viable.

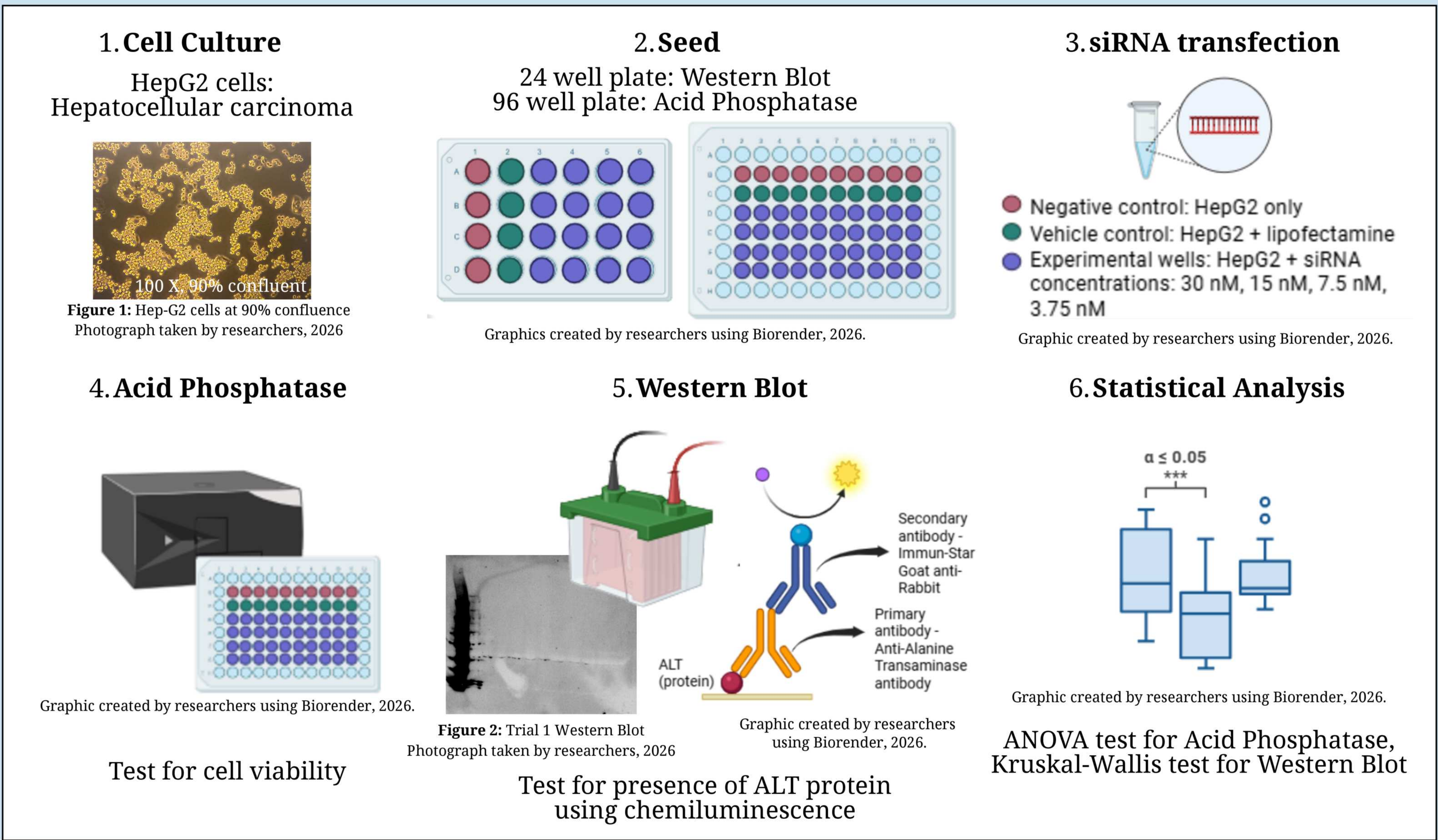
Variables

IV: GPT-targeting siRNA (30 nM, 15 nM, 7.5 nM, 3.75 nM)

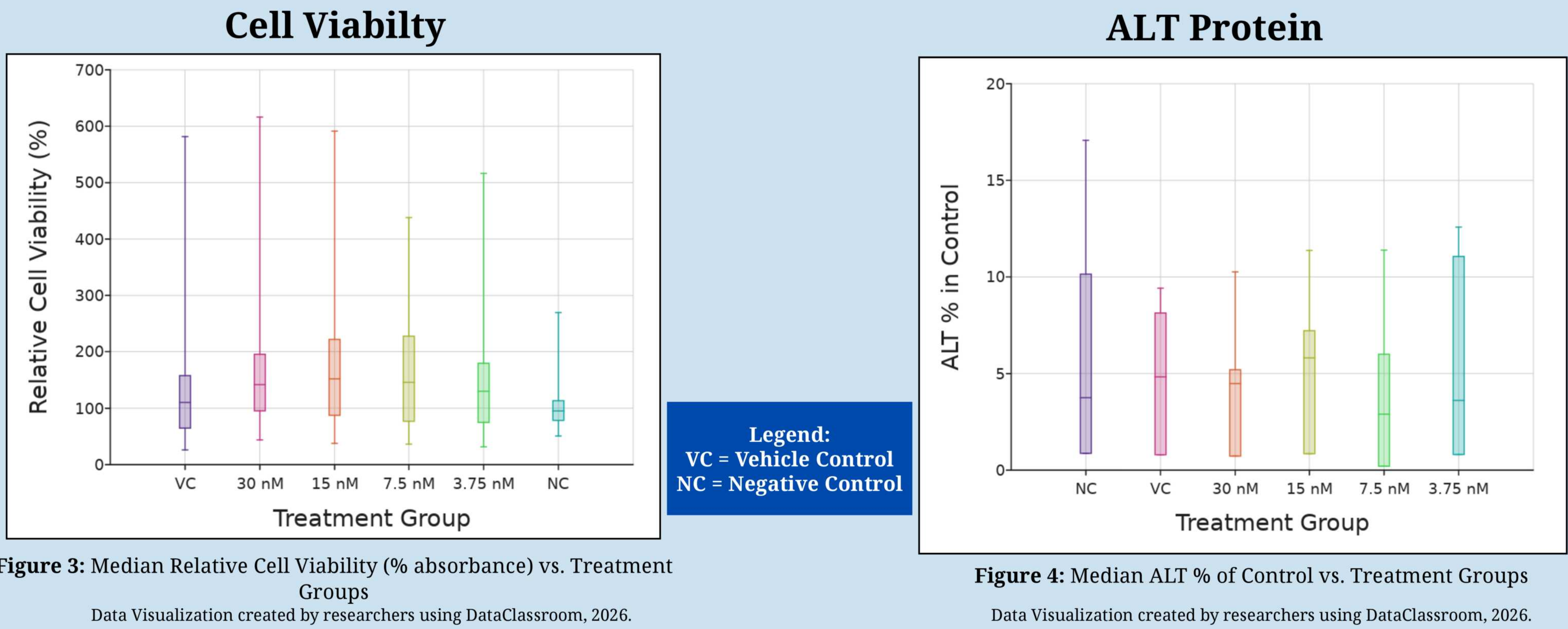
DV: ALT expression & cell viability

Controls: Vehicle (lipofectamine RNAiMAX), Negative (90% EMEM, 10% FBS)

Methodology



Results



Statistical Analysis Results

Cell Viability Post Hoc

Pairs with Significant Differences	P-values	ANOVA Test P-value
NC - VC	0.67	0.07 (A P-value of 0.07 means some confidence that the groups are different)
NC - 30 nM	0.01	
NC - 15 nM	<0.01	
NC - 7.5 nM	0.05	
NC - 3.75 nM	0.39	

Table 1: ANOVA p-values comparing relative cell viability (% absorbance) across the different treatment groups

Table created by researchers using Google Sheets, 2026.

ALT Protein Post Hoc

Pairs with Significant Differences	P-values	Kruskal-Wallis Test P-value
NC - VC	1.00	0.98 (A P-value of 0.98 means no evidence that the groups are different)
NC - 30 nM	1.00	
NC - 15 nM	1.00	
NC - 7.5 nM	1.00	
NC - 3.75 nM	1.00	

Table 2: Kruskal-Wallis p-values comparing ALT protein expression across the different treatment groups

Table created by researchers using Google Sheets, 2026.

Discussion

Data collected **did not support** hypothesis.

Statistical analysis concluded that there was **evidence to suggest there was a significant difference** between cell viability of different treatment groups and **not enough evidence to prove a significant difference** between ALT protein expression of the different groups.

Cell Viability Analysis

- Statistically significant difference in cell viability between the negative control and 30 nM, 15 nM, and 7.5 nM siRNA concentrations
 - The strongest decrease in viability correlated with higher siRNA concentrations, suggest a possible dose-dependent response.
 - Most HepG2 cells remained within a survivable range, showing that treatment did not completely diminish cellular function.
- No statistically significant difference in cell viability between negative control (NC) and vehicle control (VC)
 - Supports validity of utilizing lipofectamine RNAiMAX as siRNA transfection reagent

Conclusion:

GPT-targeting siRNA produced measurable effects on the viability of HepG2 cells, indicating that suppression of the GPT gene can influence cellular pathways in HCC.

ALT Protein Analysis

- No statistically significant difference in ALT protein levels (ALT % of control) amongst negative control (NC), vehicle control (VC), and siRNA treatment groups
 - Quantification showed fluctuations in ALT protein levels, but these changes were not consistent with the concentration curve.
- ALT protein expression remained relatively stable despite siRNA transfection and treatment
 - The lack of ALT reduction could have been due to incomplete siRNA transfection, alternative metabolic pathway activation, or variability in Western Blot quantification.

Conclusion:

GPT-targeting siRNA did not produce an identifiable decrease in ALT protein expression, although trends in cell viability suggest that GPT suppression can impact HCC cell function.

Implications

- ALT-targeting approaches need **more optimized or stronger gene-silencing methods**
- HCC treatment may require **targeting various pathways** due to complex metabolism
- siRNA-based treatments offer promising **therapeutic benefit**

Future Research

- Conduct **more trials** to compare to preliminary data
- Conduct additional trials utilizing diverse transfection methods to **improve gene knockdown efficiency**
- Use qPCR Assay to confirm GPT knockdown and **measure GPT mRNA reduction** directly
- Compare results of HepG2 cells with **other liver cancer models**

References

Alshaer, W., Zureigat, H., Al Karaki, A., Al-Kadash, A., Gharaibeh, L., Hatmal, M. M., Aljabali, A., & Awidi, A. (2021). siRNA: Mechanism of action, challenges, and therapeutic approaches. *European Journal of Pharmacology*, 905(1), 174178. <https://doi.org/10.1016/j.ejphar.2021.174178>

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